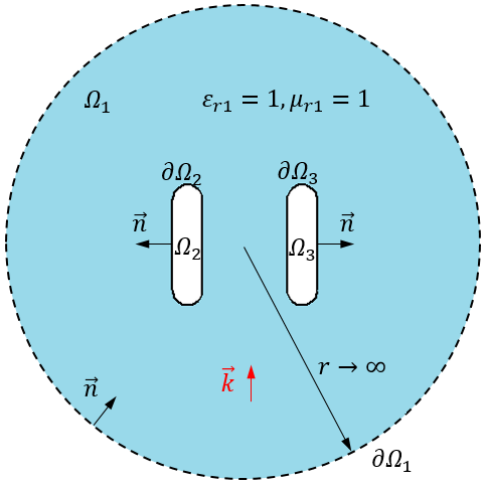
MoM for 2-D Wave Scattering Problem

The geometry of the simulation problem:



The field in the region Ω_1 can be represented as a superposition of the incident and scattered field:

$$\underline{E}_{z1}(\vec{r}) = \underline{E}_{inc}(\vec{r}) + \underline{E}_{sca}(\vec{r}) \tag{1}$$

The incident field is a plane wave:

$$\underline{E}_{inc}(\vec{r}) = \underline{E}_0\, e^{-j\vec{k}\cdot\vec{r}} \tag{2}$$

The scattered field vanishes at infinity:

$$\lim_{r\to\infty} \underline{E}_{z1}(\vec{r}) = \underline{E}_{inc}(\vec{r}) \tag{3}$$

The total field is a solution of the following BVP:

$$\Delta_{\vec{r}}\, \underline{E}_{z1}(\vec{r}) + \omega^2 \mu_1 \varepsilon_1 \underline{E}_{z1}(\vec{r}) = 0, \vec{r} \in \Omega_1 \subseteq R^2 \tag{4}$$

$$\underline{E}_{z1}(\vec{r}) = 0, \vec{r} \in \partial\Omega_2 \cup \partial\Omega_3 \subseteq R^1 \tag{5}$$

$$\underline{E}_{z1}(\vec{r}) = \underline{E}_{inc}(\vec{r}), \vec{r} \in \partial\Omega_1 \subseteq R^1 \tag{6}$$

In addition to the BVP (4) – (6) it is worth deriving the surface current density flowing at the outer surface of the conductor:

$$\vec{n} \times \underline{\vec{H}}_1 = \underline{J}_{\sigma z}\vec{e}_z, \frac{\partial \underline{E}_{z1}}{\partial n} = -j\omega\mu \underline{J}_{\sigma z}, \vec{r} \in \partial\Omega_2 \cup \partial\Omega_3 \subseteq R^1 \tag{7}$$

The Green's functions of Equation (1) is:

$$G(\vec{r}, \vec{r}\,') = -\frac{j}{4} H_0^{(2)}(k|\vec{r} - \vec{r}\,'|) \tag{*}$$

The gradient of the Green's functions can be computed as follows:

$$\nabla_{\vec{r}'} H_0^{(2)}(k|\vec{r} - \vec{r}\,'|) = \frac{dH_0^{(2)}}{dR}(kR)\, \nabla_{\vec{r}'}|\vec{r} - \vec{r}\,'| = kH_1^{(2)}(k|\vec{r} - \vec{r}\,'|)\frac{\vec{r} - \vec{r}\,'}{|\vec{r} - \vec{r}\,'|} \tag{**}$$

For the total field $\underline{E}_{z1}$ and the incident field $\underline{E}_{inc}$ it is possible to write the following integral equations:

$$C_1(\vec{r})\underline{E}_{z1}(\vec{r}) = \oint_{(\partial\Omega_{2,3}^+)} \underline{E}_{z1}(\vec{r}\,')\frac{\partial G}{\partial n_{\vec{r}\,'}}(\vec{r},\vec{r}\,')dl(\vec{r}\,') - \oint_{(\partial\Omega_{2,3}^+)} G(\vec{r},\vec{r}\,')\frac{\partial \underline{E}_{z1}}{\partial n_{\vec{r}\,'}}(\vec{r}\,')dl(\vec{r}\,') + \oint_{(\partial\Omega_1^+)} \underline{E}_{inc}(\vec{r}\,')\frac{\partial G}{\partial n_{\vec{r}\,'}}(\vec{r},\vec{r}\,')dl(\vec{r}\,') - \oint_{(\partial\Omega_1^+)} G(\vec{r},\vec{r}\,')\frac{\partial \underline{E}_{inc}}{\partial n_{\vec{r}\,'}}(\vec{r}\,')dl(\vec{r}\,') \quad (8)$$

$$\underline{E}_{inc}(\vec{r}) = \oint_{(\partial\Omega_1^+)} \underline{E}_{inc}(\vec{r}\,')\frac{\partial G}{\partial n_{\vec{r}\,'}}(\vec{r},\vec{r}\,')dl(\vec{r}\,') - \oint_{(\partial\Omega_1^+)} G(\vec{r},\vec{r}\,')\frac{\partial \underline{E}_{inc}}{\partial n_{\vec{r}\,'}}(\vec{r}\,')dl(\vec{r}\,') \quad (9)$$

where:

$$C(\vec{r}) = \begin{cases} 0, \vec{r} \notin \Omega_1 \wedge \vec{r} \notin \partial\Omega_{2,3} \\ \dfrac{1}{2}, \vec{r} \in \partial\Omega_{2,3} \\ 1, \vec{r} \in \Omega \wedge \vec{r} \notin \partial\Omega_{2,3} \end{cases} \quad (10)$$

Equations (8) and (9) can be combined into the following integral equation:

$$C_1(\vec{r})\underline{E}_{z1}(\vec{r}) = \underline{E}_{inc}(\vec{r}) + \oint_{(\partial\Omega_{2,3}^+)} \underline{E}_{z1}(\vec{r}\,')\frac{\partial G}{\partial n_{\vec{r}\,'}}(\vec{r},\vec{r}\,')dl(\vec{r}\,') - \oint_{(\partial\Omega_{2,3}^+)} G(\vec{r},\vec{r}\,')\frac{\partial \underline{E}_{z1}}{\partial n_{\vec{r}\,'}}(\vec{r}\,')dl(\vec{r}\,') \quad (11)$$

For the points $\vec{r}$ over the boundaries $\partial\Omega_{2,3}$ Equation (11) taking into account (5) can be written as follows:

$$\oint_{(\partial\Omega_{2,3}^+)} G(\vec{r},\vec{r}\,')\frac{\partial \underline{E}_{z1}}{\partial n_{\vec{r}\,'}}(\vec{r}\,')dl(\vec{r}\,') = \underline{E}_{inc}(\vec{r}) \quad (11)$$

The unknown derivative of the total field is considered constant over elements:

$$\frac{\partial \underline{E}_{z1}}{\partial n}(\vec{r}) = x_i, \vec{r} \in L_i, i = 1,2,...,N \quad (12)$$

If a finite number of line elements L_i is considered, Equation (9) becomes

$$\sum_{j=1}^{N} x_j \int_{(L_j)} G(\vec{r}_i,\vec{r}\,')dl(\vec{r}\,') = \underline{E}_{inc}(\vec{r}_i), i = 1,2,...,N \quad (13)$$

Equation (13) represents a linear system of equations:

$$[K]\{x\} = \{b\} \quad (14)$$

By solving the system (14) the unknown function $\frac{\partial \underline{E}_{z1}}{\partial n_{\vec{r}'}}$ is obtained everywhere over the boundaries. Thus, according to Equation (7), the surface current is also obtained:

$$\underline{J}_{\sigma z} = -\frac{1}{j\omega\mu}\frac{\partial \underline{E}_{z1}}{\partial n_{\vec{r}'}} \tag{15}$$

The obtained surface current density is a source of the scattered field:

$$\underline{E}_{sca}\left(\vec{r}\right) = j\omega\mu_1 \oint\limits_{\left(\partial\Omega_{2,3}^{+}\right)} G\left(\vec{r},\vec{r}\,'\right)\underline{J}_{\sigma z}\left(\vec{r}\,'\right)dl\left(\vec{r}\,'\right), \vec{r} \in \Omega_1 \tag{16}$$

The derivative of the incident field:

$$\frac{\partial \underline{E}_{inc}}{\partial n} = \nabla \underline{E}_{inc} \cdot \vec{n} = \underline{E}_0 \left(\vec{e}_x \frac{\partial}{\partial x} e^{-j\left(k_x x + k_y y\right)} + \vec{e}_y \frac{\partial}{\partial y} e^{-j\left(k_x x + k_y y\right)} \right) \cdot \vec{n} \tag{17}$$

$$\frac{\partial \underline{E}_{inc}}{\partial n} = -j\underline{E}_{inc}\vec{k}_{pw} \cdot \vec{n} \tag{18}$$